

# TRANSFORMATION OF AXES



## SYNOPSIS

- Change of axes or transformation of axes is of three types :

- i) Translation of axes
- ii) Rotation of axes
- iii) General Transformation

### Translation of axes:

- i) Shifting the origin to some other point without changing the direction of axes.  
 ii) When the origin is translated to  $(h, k)$ , the equations of transformation are  
 $x = X + h, y = Y + k$  where  $(x, y)$  are the original coordinates and  $(X, Y)$  are the new coordinates of the point.

### Rotation of axes:

- i) Rotating the system of coordinate axes through an angle ' $\theta$ ' without changing the position of the origin.  
 ii) When the axes are rotated through an angle ' $\theta$ ' in anticlockwise direction. The equations of transformation are given by

|   | X             | Y              |
|---|---------------|----------------|
| x | $\cos \theta$ | $-\sin \theta$ |
| y | $\sin \theta$ | $\cos \theta$  |

**Set-1**  $x = X \cos \theta - Y \sin \theta$   
 $y = X \sin \theta + Y \cos \theta$

**Set-2**  $X = x \cos \theta + y \sin \theta$   
 $Y = -x \sin \theta + y \cos \theta$

- Transformation is used in reducing the general equation of any curve to the desired form. For example  
 i) To eliminate first degree terms, we apply translation.  
 ii) To eliminate the term containing 'xy', we apply rotation.  
 iii) The point to which the origin has to be shifted to eliminate first degree terms (x, y terms) in  $S = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  is obtained

by solving  $\frac{\partial S}{\partial x} = 0, \frac{\partial S}{\partial y} = 0$

- iv) To remove the first degree terms from the equation  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  the origin is to be shifted to the point

$$(x_1, y_1) = \left[ \frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right], ab - h^2 \neq 0.$$

In this case, the transformed equation is  $aX^2 + 2hXY + bY^2 + (gx_1 + fy_1 + c) = 0$

**W.E-1:** When the axes are translated to the point  $(-2, 3)$  then the transformed equation of the curve  $2x^2 + 4xy + 5y^2 - 4x - 22y + 7 = 0$  is

Sol:  $a = 2, h = 2, b = 5, g = -2, f = -11, c = 7$

$$\left( \frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right) = (-2, 3) = (x_1, y_1)$$

Transformed equation is

$$aX^2 + 2hXY + bY^2 + (gx_1 + fy_1 + c) = 0$$

$$\Rightarrow 2X^2 + 4XY + 5Y^2 - 2(-2) - 11(3) + 7 = 0$$

$$\therefore 2X^2 + 4XY + 5Y^2 - 22 = 0$$

- v) To remove the first degree terms from the equation  $ax^2 + by^2 + 2gx + 2fy + c = 0$ , the

origin is to be shifted to the point  $\left( \frac{-g}{a}, \frac{-f}{b} \right)$ .

In this case, the transformed equation is

$$aX^2 + bY^2 + \left( \frac{-g^2}{a} + \frac{-f^2}{b} + c \right) = 0$$

**W.E-2:** When the axes are translated to the point  $(-1, 1)$  then the transformed equation of the curve  $x^2 + 2y^2 + 2x - 4y + 2 = 0$  is

Sol:  $a = 1, b = 2, g = 1, f = -2, c = 2$

$$\text{new origin} = \left( \frac{-g}{a}, \frac{-f}{b} \right) = (-1, 1)$$

Transformed equation is

$$aX^2 + bY^2 + \left( \frac{-g^2}{a} + \frac{-f^2}{b} + c \right) = 0$$

$$\Rightarrow X^2 + 2Y^2 + (-1 - 2 + 2) = 0$$

$$\therefore X^2 + 2Y^2 = 1$$

vi) To remove the first degree terms from  $2hxy + 2gx + 2fy + c = 0$ , the origin is to be

shifted to the point  $\left(\frac{-f}{h}, \frac{-g}{h}\right)$ . In this case,

the transformed equation is

$$2h^2XY - 2gf + ch = 0$$

**W.E-3 :** When the origin is shifted to the point (5, -2) then the transformed equation of the curve  $xy + 2x - 5y + 11 = 0$  is

Sol:  $h = \frac{1}{2}, g = 1, f = -\frac{5}{2}, c = 11$

$$\left(\frac{-f}{h}, \frac{-g}{h}\right) = (5, -2)$$

Transformed equation is  $2h^2XY - 2fg + ch = 0$

$$\therefore XY = -21$$

vii) The point to which the origin has to be shifted to eliminate x and y terms in the equation

$$a(x + \alpha)^2 + b(y + \beta)^2 = c \text{ is } (-\alpha, -\beta)$$

**W.E-4 :** The point to which the origin has to be shifted to eliminate x and y terms in the equation  $2x^2 + 3y^2 - 12x - 12y + 21 = 0$  is

Sol: First method :

$$2x^2 + 3y^2 - 12x - 12y + 21 = 0$$

$$a = 2, b = 3, g = -6, f = -6$$

$$\therefore \text{New origin} = \left(-\frac{g}{a}, -\frac{f}{b}\right) = (3, 2)$$

Second method :

$$2x^2 + 3y^2 - 12x - 12y + 21 = 0$$

$$\Rightarrow 2(x^2 - 6x) + 3(y^2 - 4y) + 21 = 0$$

$$\Rightarrow 2(x^2 - 6x + 9) + 3(y^2 - 4y + 4) - 18 - 12 + 21 = 0$$

$$\Rightarrow 2(x - 3)^2 + 3(y - 2)^2 = 9$$

$$\text{Comparing with } a(x + \alpha)^2 + b(y + \beta)^2 = c,$$

we get  $\alpha = -3, \beta = -2$

$$\therefore \text{New Origin} = (-\alpha, -\beta) = (3, 2)$$

viii) a) To remove xy term of

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ the}$$

angle of rotation of axes is

$$\theta = \frac{1}{2} \tan^{-1} \left( \frac{2h}{a-b} \right), \text{ if } a \neq b$$

$$= (2n+1)\frac{\pi}{4}, n \in \mathbb{Z} \text{ if } a = b$$

b) If ' $\theta$ ' is angle of rotation to eliminate XY term

in  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ , then

$n\frac{\pi}{2} + \theta$ ,  $n \in \mathbb{Z}$  is also an angle of rotation to

eliminate XY term

ix) The angle of rotation of axes so that the equation  $ax^2 + by^2 + c = 0$  is reduced as

a) X = constant is  $\tan^{-1} \left( \frac{b}{a} \right)$

b) Y = constant is  $\tan^{-1} \left( -\frac{a}{b} \right)$

**W.E.-5 :** The angle of rotation of the axes so that

the equation  $\sqrt{3}x - y + 5 = 0$  may be reduced to the form  $y = \text{constant}$  is

Sol:  $\sqrt{3}x - y + 5 = 0$

$$a = \sqrt{3}, b = -1$$

$$\therefore \theta = \tan^{-1} \left( -\frac{a}{b} \right) = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$$

x) The equation

$S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  has trans-

formed to  $AX^2 + 2HXY + BY^2 + 2GX + 2FY +$

$C = 0$ , when the origin is shifted to  $(l, m)$  then

$$A = a; B = b; H = h;$$

$$2G = \left( \frac{\partial S}{\partial x} \right)_{(l,m)} \quad 2F = \left( \frac{\partial S}{\partial y} \right)_{(l,m)} \quad C = S(l, m)$$

➤ The condition that the equation  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  to take the form

$aX^2 + 2hXY + bY^2 = 0$  when the axes are

translated is  $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

**General Transformation:**

➤ i) Applying both translation and rotation.

ii) The equations of general transformation are given by

|         | X             | Y              |
|---------|---------------|----------------|
| $x - h$ | $\cos \theta$ | $-\sin \theta$ |
| $y - k$ | $\sin \theta$ | $\cos \theta$  |

**Note :** 1) If the rotation is in clockwise direction then replace  $\theta$  by  $-\theta$ .

2) On translation or rotation the position of the point, length of line segment, area, perimeter, angles are not changed. But the coordinates and equations will change.

3) If the axes are rotated through an angle  $\theta$  the following are the some invariants in original and transformed equations of second degree general equation.

- 1)  $a + b$     2)  $h^2 - ab$     3)  $c$   
 4)  $(a - b)^2 + 4h^2$     5)  $2fgh - af^2 - bg^2$   
 6)  $abc + 2fgh - af^2 - bg^2 - ch^2$

➤ **If the Origin is shifted at  $O'(h, k)$  and the axis are rotated about the new origin  $O'$  by an angle  $\theta$  in anticlock wise sense, then**  
 $x = h + X \cos \theta - Y \sin \theta, y = k + X \sin \theta + Y \cos \theta$

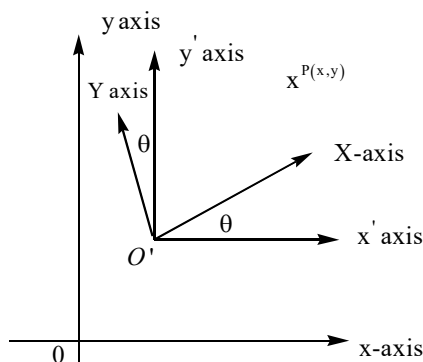
**Proof :** If  $O$  is shifted to  $O'(h, k)$  then new

**co-ordinates of  $P(x, y)$  become  $(x^1, y^1)$**

$$x = h + x^1, y = k + y^1$$

**Axis are rotated through an angle  $\theta$ , in the anticlockwise sense than the new**

**co-ordinates of  $P(x^1, y^1)$  become  $(X, Y)$**



$$\Rightarrow x^1 = X \cos \theta - Y \sin \theta, y^1 = X \sin \theta + Y \cos \theta$$

$$\therefore x = h + X \cos \theta - Y \sin \theta, y = k + X \sin \theta + Y \cos \theta$$

➤ **If by change of axes without change of origin the expression  $ax^2 + 2hxy + by^2$  becomes  $a_1X^2 + 2h_1XY + b_1Y^2$  then**

i)  $a + b = a_1 + b_1$

ii)  $ab - h^2 = a_1b_1 - h_1^2$

iii)  $(a - b)^2 + 4h^2 = (a_1 - b_1)^2 + 4h_1^2$

**Proof :**

**Let the axes be rotated through an angle  $\alpha$  in anticlockwise sense**

$$\Rightarrow x = X \cos \alpha - Y \sin \alpha, y = X \sin \alpha + Y \cos \alpha$$

$$\begin{aligned} ax^2 + 2hxy + by^2 &= a(X \cos \alpha - Y \sin \alpha)^2 \\ &+ 2h(X \cos \alpha - X \sin \alpha)(X \sin \alpha + Y \cos \alpha) + \\ &b(X \sin \alpha + Y \cos \alpha)^2 \end{aligned}$$

$$\text{But } ax^2 + 2hxy + by^2$$

$$\text{transformed into } a_1X^2 + 2h_1XY + b_1Y^2$$

$$\Rightarrow a_1 = a \cos^2 \alpha + h \sin 2\alpha + b \sin^2 \alpha$$

$$2h_1 = 2h \cos 2\alpha - a \sin 2\alpha + b \sin 2\alpha$$

$$b_1 = a \sin^2 \alpha - h \sin 2\alpha + b \cos^2 \alpha$$

**by using these conditions, i)  $a + b = a_1 + b_1$**

ii)  $ab - h^2 = a_1b_1 - h_1^2$

iii)  $(a - b)^2 + 4h^2 = (a_1 - b_1)^2 + 4h_1^2$



### LEVEL - I (C.W)

#### Translation of axes :

- If (2, 3) are the coordinates of a point P in the new system when the origin is shifted to (-3, 7) then the original coordinates of P are  
 1) (-1, 10)    2) (5, -4)    3) (-5, 4)    4) (-1, 5)
- The coordinates of the point (4, 5) in the new system, when its origin is shifted to (3, 7) are  
 1) (1, 2)    2) (-1, 2)    3) (-1, -2)    4) (1, -2)
- When the origin is shifted to a point P, the point (2, 0) is transformed to (0, 4) then the coordinates of P are

- 1) (2, -4) 2) (-2, 4) 3) (-2, -4) 4) (2, 4)
4. If the axes are translated to the point (-2, -3) then the equation  $x^2 + 3y^2 + 4x + 18y + 30 = 0$  transforms to
- 1)  $X^2 + Y^2 = 4$  2)  $X^2 + 3Y^2 = 1$   
 3)  $X^2 - Y^2 = 4$  4)  $X^2 - 3Y^2 = 1$
5. When the axes are translated to the point (5, -2) then the transformed form of the equation  $xy + 2x - 5y - 11 = 0$  is
- 1)  $\frac{X}{Y} = 1$  2)  $\frac{Y}{X} = 1$  3)  $XY = 1$  4)  $XY^2 = 2$
6. If the transformed equation of a curve when the origin is translated to (1, 1) is  $X^2 + Y^2 + 2X - Y + 2 = 0$  then the original equation of the curve is
- 1)  $x^2 + 2y^2 = 1$   
 2)  $x^2 + y^2 + 3y + 3 = 0$   
 3)  $x^2 + y^2 + 3y - 3 = 0$   
 4)  $x^2 + y^2 - 3y + 3 = 0$
7. In order to make the first degree terms missing in the equation  $2x^2 + 7y^2 + 8x - 14y + 15 = 0$ , the origin should be shifted to the point
- 1) (1, -2) 2) (-2, -1) 3) (2, 1) 4) (-2, 1)
8. The point to which the origin should be shifted in order to remove the x and y terms in the equation  $14x^2 - 4xy + 11y^2 - 36x + 48y + 41 = 0$  is
- 1) (1, -2) 2) (-2, 1) 3) (-1, 2) 4) (2, -1)
9. If the distance between the two given points is 2 units and the points are transferred by shifting the origin to (2, 2), then the distance between the points in their new position is
- 1) 2 2) 5 3) 6 4) 7
10. When (0, 0) shifted to (3, -3) the coordinates of P(5, 5), Q(-2, 4) and R(7, -7) in the new system are A, B, C then area of triangle ABC in sq units is
- 1) 43 2) 23 3) 45 4) 50
- Rotation of axes :**
11. When axes are rotated through an angle of  $45^\circ$  in positive direction without chang-

ing origin then the coordinates of  $(\sqrt{2}, 4)$  in old system are

- 1)  $(1 - 2\sqrt{2}, 1 + 2\sqrt{2})$  2)  $(1 + 2\sqrt{2}, 1 - 2\sqrt{2})$   
 3)  $(2\sqrt{2}, \sqrt{2})$  4)  $(2, \sqrt{2})$
12. If the axes are rotated through an angle  $30^\circ$ , the coordinates of  $(2\sqrt{3}, -3)$  in the new system are
- 1)  $\left(\frac{3}{2}, \frac{-5}{2}\right)$  2)  $\left(\frac{-\sqrt{3}}{2}, \frac{5}{2}\right)$   
 3)  $\left(\frac{3}{2}, \frac{-5\sqrt{3}}{2}\right)$  4)  $\left(3\sqrt{2}, \frac{-5\sqrt{3}}{2}\right)$
13. The transformed equation of  $x^2 + 2\sqrt{3}xy - y^2 - 8 = 0$ , when the axes are rotated through an angle  $\frac{\pi}{6}$  is
- 1)  $X^2 - Y^2 = 0$  2)  $X^2 - Y^2 = 4$   
 3)  $X^2 - Y^2 = 2$  4)  $X^2 + Y^2 = 4$
14. If the axes are rotated through an angle  $180^\circ$  then the equation  $2x - 3y + 4 = 0$  becomes
- 1)  $2X - 3Y - 4 = 0$  2)  $2X + 3Y - 4 = 0$   
 3)  $3X - 2Y + 4 = 0$  4)  $3X + 2Y + 4 = 0$
15. If the transformed equation of a curve is  $17X^2 - 16XY + 17Y^2 = 225$  when the axes are rotated through an angle  $45^\circ$ , then the original equation of the curve is
- 1)  $25x^2 + 9y^2 = 225$  2)  $9x^2 + 25y^2 = 225$   
 3)  $25x^2 - 9y^2 = 225$  4)  $9x^2 - 25y^2 = 225$
16. If the equation  $4x^2 + 2\sqrt{3}xy + 2y^2 - 1 = 0$  becomes  $5X^2 + Y^2 = 1$ , when the axes are rotated through an angle  $\theta$ , then  $\theta$  is
- 1)  $15^\circ$  2)  $30^\circ$  3)  $45^\circ$  4)  $60^\circ$
17. The angle of rotation of axes in order to eliminate xy term in the equation  $xy = c^2$  is
- 1)  $\frac{\pi}{12}$  2)  $\frac{\pi}{6}$  3)  $\frac{\pi}{3}$  4)  $\frac{\pi}{4}$
18. The transformed equation of  $x^2 + y^2 = r^2$ , when the axes are rotated through an

angle  $36^\circ$  is

- 1)  $\sqrt{5}X^2 - 4XY + Y^2 = r^2$
- 2)  $X^2 + 2XY - \sqrt{5}Y^2 = r^2$
- 3)  $X^2 - Y^2 = r^2$
- 4)  $X^2 + Y^2 = r^2$

**LEVEL-I (C.W)-KEY**

- 1) 1    2) 4    3) 1    4) 2    5) 3
- 6) 4    7) 4    8) 1    9) 1    10) 1
- 11) 1    12) 3    13) 2    14) 1    15) 1
- 16) 2    17) 4    18) 4

**LEVEL-I (C.W)-HINTS**

1.  $(X, Y) = (2, 3)$   
 $(h, k) = (-3, 7)$   
 $(x, y) = (X+h, Y+k)$
2.  $(x, y) = (4, 5)$   
 $(h, k) = (3, 7)$   
 $(X, Y) = (x-h, y-k)$
3.  $(x, y) = (2, 0)$   
 $(X, Y) = (0, 4)$   
 New origin  $(x-X, y-Y)$
4.  $(h, k) = (-2, -3)$   
 Put  $x = X - 2, y = Y - 3$   
 $X^2 + 3Y^2 = 1$
5.  $(h, k) = (5, -2)$   
 Put  $x = X + 5, y = Y - 2$   
 $\Rightarrow XY = 1$
6. put  $X = x-1, Y = y-1$
7.  $a = 2, b = 7, g = 4, f = -7, c = 15$   
 New origin  $= \left( \frac{-g}{a}, \frac{-f}{b} \right) = (-2, 1)$
8. Use  $\left( \frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$
9. Distance remains same
10. Area of triangle ABC = Area of triangle PQR
11. Use  $x = X \cos \theta - Y \sin \theta$

$$y = X \sin \theta + Y \cos \theta$$

12. Use  $X = x \cos \theta + y \sin \theta$   
 $Y = -x \sin \theta + y \cos \theta$
13. Use  $x = X \cos \theta - Y \sin \theta$   
 $y = X \sin \theta + Y \cos \theta$
14.  $x = X \cos 180^\circ - Y \sin 180^\circ,$   
 $y = X \sin 180^\circ + Y \cos 180^\circ,$
15.  $X = x \cos 45^\circ + y \sin 45^\circ,$   
 $Y = -x \sin 45^\circ + y \cos 45^\circ,$
16.  $\theta = \frac{1}{2} \tan^{-1} \left( \frac{2h}{a-b} \right) = 30^\circ$
17.  $\theta = \frac{1}{2} \tan^{-1} \left( \frac{2h}{a-b} \right)$
18.  $x = X \cos 36^\circ - Y \sin 36^\circ,$   
 $y = X \sin 36^\circ + Y \cos 36^\circ$



**LEVEL-I (H.W)**

**Translation of axes :**

1. If  $(3, 2)$  are coordinates of a point 'P' in the new system when origin is shifted to  $(3, 7)$ , then the original coordinates of 'P' are  
 1)  $(6, 9)$     2)  $(-6, 9)$     3)  $(6, -9)$     4)  $(6, 0)$
2. The coordinates of the point  $(3, 7)$  in the new system when the origin is shifted to  $(5, -1)$  are  
 1)  $(8, 6)$     2)  $(2, -8)$     3)  $(-2, 8)$     4)  $(4, 3)$
3. If the point  $(5, 7)$  is transformed to  $(-1, 2)$  when the origin is shifted to A, then A =  
 1)  $(4, 9)$     2)  $(6, 5)$     3)  $(-6, -5)$     4)  $(2, 4)$
4. If the origin is shifted to the point  $(-1, 2)$  without changing the direction of axes, the equation  $x^2 - y^2 + 2x + 4y = 0$  becomes  
 1)  $X^2 + Y^2 + 3 = 0$     2)  $X^2 + Y^2 - 3 = 0$   
 3)  $X^2 - Y^2 + 3 = 0$     4)  $X^2 - Y^2 - 3 = 0$
5. When the axes are translated to the point

$\left(1, \frac{1}{2}\right)$ , the equation

$5x^2 + 4xy + 8y^2 - 12x - 12y = 0$  transforms to

1)  $5X^2 + 4XY + 8Y^2 = 9$

2)  $2X^2 + 3XY + 4Y^2 = 0$

3)  $X^2 + 2Y^2 - 3Y = 0$

4)  $X^2 - 7XY + 8Y^2 = 0$

6. If (3,-4) is the point to which the origin is shifted and the transformed equation is  $X^2 + Y^2 = 4$  then the original equation is

1)  $x^2 + y^2 + 6x + 8y + 21 = 0$

2)  $x^2 + y^2 + 6x + 8y - 21 = 0$

3)  $x^2 + y^2 - 6x + 8y + 21 = 0$

4)  $x^2 + y^2 - 6x - 8y + 21 = 0$

7. The point to which the origin should be shifted in order to eliminate x and y terms in the equation  $x^2 + 3y^2 - 2x + 12y + 1 = 0$  is

1) (1,-2) 2) (1,3) 3) (-4,3) 4) (-1,2)

8. The point to which origin is shifted in order to miss the first degree terms in

$2x^2 + 5xy + 3y^2 + 6x + 7y + 1 = 0$  is

1) (2,1) 2) (1,-2) 3) (2,-1) 4) (1,2)

9. If the area of a triangle is 5 sq. units, then the area of the triangle when the origin is shifted to (1,2) is

1) 2 sq.u 2) 3 sq.u 3) 4 sq.u 4) 5 sq.u

### Rotation of axes :

10. When axes are rotated by an angle of  $135^\circ$  initial coordinates of (4,-3) are

1)  $\left(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$  2)  $\left(\frac{1}{\sqrt{2}}, \frac{-7}{\sqrt{2}}\right)$

3)  $\left(\frac{-1}{\sqrt{2}}, \frac{-7}{\sqrt{2}}\right)$  4)  $\left(\frac{-1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$

11. If the axes are rotated through an angle  $30^\circ$  in the clockwise direction, the point  $(4, 2\sqrt{3})$  in the new system is

1) (2,3) 2)  $(2, \sqrt{3})$  3)  $(\sqrt{3}, 2)$  4)  $(\sqrt{3}, 5)$

12. The transformed equation of  $3x^2 + 3y^2 + 2xy = 2$  when the coordinate axes are rotated through an angle of  $45^\circ$  is (EAMCET - 2008)

1)  $X^2 + 2Y^2 = 1$  2)  $2X^2 + Y^2 = 1$

3)  $X^2 + Y^2 = 1$  4)  $X^2 + 3Y^2 = 1$

13. When the axes are rotated through an angle  $90^\circ$  the equation  $5x - 2y + 7 = 0$  transforms to

1)  $2X - 5Y + 7 = 0$  2)  $2X + 5Y - 7 = 0$

3)  $2X - 5Y - 7 = 0$  4)  $2X + 5Y + 7 = 0$

14. If the transformed equation of a curve is  $9X^2 + 16Y^2 = 144$  when the axes are rotated through an angle of  $45^\circ$  then the original equation of a curve is

1)  $25x^2 + 14xy + 25y^2 = 288$

2)  $25x^2 + 14xy - 25y^2 = 288$

3)  $25x^2 - 14xy + 25y^2 = 288$

4)  $25x^2 - 14xy - 25y^2 = 288$

15. The angle of rotation of axes in order to eliminate xy term in the equation

$x^2 + 2\sqrt{3}xy - y^2 = 2a^2$  is

1)  $\pi/6$  2)  $\pi/4$  3)  $\pi/3$  4)  $\pi/2$

16. The transformed equation of  $x\cos\alpha + y\sin\alpha = P$  when the axes are rotated through an angle  $\alpha$  is

1)  $X = P$  2)  $X + P = 0$

3)  $Y = P$  4)  $Y + P = 0$

### LEVEL-I (H.W)- KEY

1) 1 2) 3 3) 2 4) 3 5) 1  
6) 3 7) 1 8) 2 9) 4 10) 4  
11) 4 12) 2 13) 2 14) 3 15) 1  
16) 1

### LEVEL-I (H.W)- HINTS

1.  $(X,Y)=(3,2), (h,k)=(3,7), (x,y)=(X+h,Y+k)$   
2.  $(x,y)=(3,7), (h,k)=(5,-1)$

- $(X, Y) = (x - h, y - k)$
- $(x, y) = (5, 7), (X, Y) = (-1, 2)$   
 $A = (x - X, y - Y) = (6, 5)$
  - $(h, k) = (-1, 2)$  Put  $x = X - 1, y = Y + 2$  transformed equation is  $X^2 - Y^2 + 3 = 0$
  - $(h, k) = (1, 1/2)$   
 put  $x = X + 1, y = Y + \frac{1}{2}$  in given equation
  - Put  $X = x - 3, Y = y + 4$
  - Use  $\left(\frac{-g}{a}, \frac{-f}{b}\right)$
  - Use  $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right)$
  - Area remains the same
  - $X = x \cos \theta + y \sin \theta$   
 $Y = -x \sin \theta + y \cos \theta$
  - $X = x \cos \theta + y \sin \theta$   
 $Y = -x \sin \theta + y \cos \theta$   
 Where  $\theta = -30^\circ$
  - $x = X \cos \theta - Y \sin \theta$   
 $y = X \sin \theta + Y \cos \theta$
  - $x = X \cos 90^\circ - Y \sin 90^\circ$   
 $y = X \sin 90^\circ + Y \cos 90^\circ$
  - Use  $X = x \cos 45^\circ - y \sin 45^\circ$   
 $Y = -x \sin 45^\circ + y \cos 45^\circ$
  - $\theta = \frac{1}{2} \tan^{-1} \left( \frac{2h}{a - b} \right)$
  - Use  $x = X \cos \alpha - Y \sin \alpha$   
 $y = X \sin \alpha + Y \cos \alpha$



**LEVEL-II (C.W)**

**Translation of axes :**

- To remove the first degree terms of the equation  $2xy + 4x - 2y + 7 = 0$  the shifted origin

is

- 1)  $(2, -1)$  2)  $(-1, 2)$  3)  $(1, -2)$  4)  $(-2, 1)$
- By translating the axes the equation  $xy - 2x - 3y - 4 = 0$  has changed to  $XY = k$ , then  $k =$   
 1)  $-10$  2)  $10$  3)  $4$  4)  $-4$
- The origin is shifted to  $(1, 2)$ , the equation  $y^2 - 8x - 4y + 12 = 0$  changes to  $Y^2 + 4aX = 0$  then  $a =$   
 1)  $2$  2)  $-2$  3)  $1$  4)  $-1$
- When the origin is shifted to a suitable point, the equation  $2x^2 + y^2 - 4x + 4y = 0$  transformed as  $2X^2 + Y^2 - 8X + 8Y + 18 = 0$ . The point to which origin was shifted is  
 1)  $(1, 2)$  2)  $(1, -2)$  3)  $(-1, 2)$  4)  $(-1, -2)$
- If  $(\cos \alpha, \cos \beta)$  are the new co-ordinates of a point P when the axes are translated to the point  $(1, 1)$ , then the original coordinates are  
 1)  $(2 \cos^2 \alpha/2, 2 \cos^2 \beta/2)$   
 2)  $(2 \cos^2 \alpha/2, 2 \sin^2 \beta/2)$   
 3)  $(2 \sin^2 \alpha/2, 2 \cos^2 \beta/2)$   
 4)  $(-2 \cos^2 \alpha/2, -2 \cos^2 \beta/2)$
- The first degree terms of  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  are removed by shifting origin to  $(\alpha, \beta)$ . The new equation is  
 1)  $ax^2 + 2hxy + by^2 + 2y\alpha + 2b\beta + c = 0$   
 2)  $ax^2 + 2hxy + by^2 + g\alpha + f\beta + c = 0$   
 3)  $ax^2 + 2hxy + by^2 + h\alpha + b\beta + c = 0$   
 4)  $ax^2 + 2hxy - by^2 - h\alpha - b\beta - c = 0$
- When  $(0, 0)$  shifted to  $(2, -2)$  the transformed equation of  $(x - 2)^2 + (y + 2)^2 = 9$  is  
 1)  $X^2 + Y^2 = 9$   
 2)  $X^2 + 3Y^2 = 9$   
 3)  $X^2 + Y^2 - 2X + 6Y = 0$   
 4)  $4X^2 + 9Y^2 = 36$

**Rotation of axes :**

8. The transformed equation of  $4xy - 3x^2 = 10$  when the axes are rotated through an angle whose tangent is '2' is

1)  $X^2 - 4Y^2 = 10$       2)  $4X^2 - Y^2 = 10$   
 3)  $XY - 10 = 0$       4)  $2X^2 - Y^2 + 10 = 0$

9. The angle of rotation of the axes so that the equation  $\sqrt{3}x - y + 5 = 0$  may be reduced to the form  $Y = k$ , where  $k$  is a constant is

1)  $\pi/6$     2)  $\pi/4$     3)  $\pi/3$     4)  $\pi/12$

10. The angle of rotation of the axes so that the equation  $ax + by + c = 0$  may be reduced to  $X = p$  is

1)  $\tan^{-1} \frac{b}{a}$     2)  $\tan^{-1} \frac{a}{b}$     3)  $\frac{\pi}{2}$     4)  $\frac{\pi}{3}$

11. The coordinate axes are rotated about the origin 'O' in the counter clockwise direction through an angle  $60^\circ$ . If  $a$  and  $b$  are the intercepts made on the new axes by a straight line whose equation referred to the

original axes is  $3x + 4y - 5 = 0$  then  $\frac{1}{a^2} + \frac{1}{b^2} =$

1)  $1/25$     2)  $1/9$     3)  $1/16$     4)  $1$

12. The coordinate axes are rotated through an angle  $\theta$  about the origin in anticlock-wise sense. If the equation  $2x^2 + 3xy - 6x + 2y - 4 = 0$  changes to  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  then  $a+b$  is equal to

1)  $3 \cos \theta - 3 \sin \theta$     2)  $3 \cos \theta + 2 \sin \theta$   
 3)  $1$     4)  $2$

13. Let  $L$  be the line  $2x + y - 2 = 0$ . The axes are rotated by  $45^\circ$  in clockwise direction then the intercepts made by the line  $L$  on the new axes are respectively

1)  $1, \sqrt{2}$     2)  $\sqrt{2}, 1$   
 3)  $2\sqrt{2}, \frac{2\sqrt{2}}{3}$     4)  $\frac{2\sqrt{2}}{3}, 2\sqrt{2}$

14. A point  $(2, 2)$  undergoes reflection in the  $x$ -axis and then the coordinate axes are rotated through an angle of  $\pi/4$  in anticlockwise direction. The final position of the point in the new coordinate system is

1)  $(0, 2\sqrt{2})$     2)  $(0, -2\sqrt{2})$

3)  $(2\sqrt{2}, 0)$     4)  $(-2\sqrt{2}, 0)$

15. The acute angle  $\theta$  through which the coordinate axes should be rotated for the point  $A(2, 4)$  to attain the new abscissa 4 is given by

1)  $\tan \theta = 3/4$     2)  $\tan \theta = 5/6$   
 3)  $\tan \theta = 7/8$     4)  $\tan \theta = \frac{3}{2}$

16. A line has intercepts  $a, b$  on the axes when the axes are rotated through an angle  $\alpha$ , the line makes equal intercepts on axes then  $\tan \alpha =$

1)  $\left(\frac{a+b}{a-b}\right)$     2)  $\left(\frac{a-b}{a+b}\right)$     3)  $\left(\frac{a}{b}\right)$     4)  $\left(\frac{b}{a}\right)$

17. The new equation of the curve  $4(x - 2y + 1)^2 + 9(2x + y + 2)^2 = 25$ , if the lines  $2x + y + 2 = 0$  and  $x - 2y + 1 = 0$  are taken as the new  $x$  and  $y$  axes respectively is

1)  $4X^2 + 9Y^2 = 5$     2)  $4X^2 + 9Y^2 = 25$   
 3)  $4X^2 + 9Y^2 = 7$     4)  $4X^2 - 9Y^2 = 7$

18. The line joining the points  $A(2, 0)$  and  $B(3, 1)$  is rotated through an angle of  $45^\circ$ , about  $A$  in the anticlock wise direction. the coordinates of  $B$  in the new position

(EAMCET 2011)

1)  $(2, \sqrt{2})$     2)  $(\sqrt{2}, 2)$     3)  $(2, 2)$     4)  $(\sqrt{2}, \sqrt{2})$

19. If the axes are translated to the circumcentre of the triangle formed by  $(9, 3), (-1, 7), (-1, 3)$ , then the centroid of the triangle in the new system is

1)  $\left(5, \frac{5}{3}\right)$     2)  $(4, 3)$

3)  $\left(\frac{-5}{3}, \frac{-2}{3}\right)$     4)  $(0, 0)$

20. The point  $(4, 1)$  undergoes the following two transformations successively  
 (i) reflection about the line  $y = x$   
 (ii) translation through a distance 2 units along the positive direction of  $x$ -axis. The final position of the point is

1)  $(3, 4)$     2)  $(4, 3)$

- 3)  $(-1, 4)$                       4)  $(1, 4)$

**LEVEL-II (C.W)- KEY**

- 1) 3    2) 2    3) 2    4) 3    5) 1  
6) 2    7) 1    8) 1    9) 3    10) 1  
11) 4    12) 4    13) 3    14) 2    15) 1  
16) 2    17) 1    18) 1    19) 3    20) 1

**LEVEL-II (C.W)- HINTS**

- New origin =  $\left(\frac{-f}{h}, \frac{-g}{h}\right)$
- New origin =  $\left(\frac{-f}{h}, \frac{-g}{h}\right) = (3, 2)$   
Put  $x = X+3, y = Y+2$
- $(h, k) = (1, 2)$   
 $x = X+1, y = Y+2$   
 $Y^2 - 8X = 0$   
 $\therefore a = -2$
- Sub  $x = X+h, y = Y+k$  in  
 $2x^2 + y^2 - 4x + 4y = 0$
- $(x, y) = (X+h, Y+k) = \left(2\cos^2\frac{\alpha}{2}, 2\cos^2\frac{\beta}{2}\right)$
- $x = X+\alpha, y = Y+\beta$
- $(h, k) = (2, -2)$
- $\tan\theta = 2$ , Find  $\sin\theta$  and  $\cos\theta$
- use  $\theta = \tan^{-1}\left(\frac{-a}{b}\right)$
- $a(X\cos\theta - Y\sin\theta) + b(X\sin\theta + Y\cos\theta) + c = 0$   
coefficient of  $Y=0$
- $p = \frac{5}{3}, q = \frac{5}{4}, \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$
- $a+b = 2+0=2$
- $\theta = -45^\circ$
- Reflection of  $(2, 2)$  in X-axis is  $(2, -2) = (x, y)$   
use  $X = x\cos\theta + y\sin\theta$ ,  
 $Y = -x\sin\theta + y\cos\theta$
- $(x, y) = (2, 4), X = 4$   
Using  $X = x\cos\theta + y\sin\theta$   
 $\Rightarrow 2\cos\theta + 4\sin\theta = 4$ , dividing by  $\sin\theta$   
 $\Rightarrow 2 + 4\tan\theta = 4\sec\theta$ ,

s.b.s  $\therefore \tan\theta = \frac{3}{4}$

- $\frac{x}{a} + \frac{y}{b} = 1$   
 $\frac{(X\cos\alpha - Y\sin\alpha)}{a} + \frac{(X\sin\alpha + Y\cos\alpha)}{b} = 1$   
x.coefficient = y coefficient  
 $(a-b)\cos\alpha = (a+b)\sin\alpha$   
 $\Rightarrow \tan\alpha = \frac{a-b}{a+b}$
- Take  
 $4\left(\frac{x-2y+1}{\sqrt{5}}\right)^2 + 9\left(\frac{2x+y+2}{\sqrt{5}}\right)^2 = 5$   
 $\therefore 4X^2 + 9Y^2 = 5$
- $AB = \sqrt{1+1} = \sqrt{2}$   
 $\therefore$  by distance, verification the new coordinates of B are  $(2, \sqrt{2})$
- Given points forms a right angle triangle.  
circum centre = Mid point of AB =  $(4, 5)$   
Centroid  $G = \left(\frac{7}{3}, \frac{13}{3}\right)$   
 $\therefore$  centroid in the new system  
 $= \left(\frac{7}{3} - 4, \frac{13}{3} - 5\right) = \left(\frac{-5}{3}, \frac{-2}{3}\right)$
- Reflection of  $(4, 1)$  with reference to  $y = x$  is  
 $(1, 4)$ .  
The new coordinates of  
 $(1, 4)$  is  $(1+2, 4) = (3, 4)$



**LEVEL-II (H.W)**

**Translation of axes :**

- The point to which the origin should be translated in order to make the first degree terms missing in the equation  
 $3xy - 2x + y - 8 = 0$  is  
1)  $\left(-\frac{1}{3}, \frac{2}{3}\right)$                       2)  $\left(-\frac{1}{3}, -\frac{2}{3}\right)$

$$3) \left( \frac{2}{3}, -\frac{1}{3} \right) \quad 4) \left( -\frac{2}{3}, \frac{1}{3} \right)$$

2. By translation of axes the equation  $xy - x + 2y - 6 = 0$  changed as  $XY = c$  then  $c =$   
 1) 4      2) 5      3) 6      4) 7
3. By shifting origin to  $(-1, 2)$  the equation  $y^2 + 8x - 4y + 12 = 0$  changes as  $Y^2 = 4aX$  then  $a =$   
 1) 1      2) 2      3) -2      4) -1
4. The transformed equation of  $4x^2 + 9y^2 - 8x + 36y + 4 = 0$  when the axes are translated to  $(1, -2)$  is  $aX^2 + bY^2 = c$ . Then descending order of  $a, b, c$   
 1)  $c, b, a$     2)  $c, a, b$     3)  $a, b, c$     4)  $a, c, b$
5. On shifting the origin to a particular point, the equation  $x^2 + y^2 - 4x - 6y - 12 = 0$  transforms to  $X^2 + Y^2 = K$ . Then  $K =$   
 1) 12      2) 25      3) 24      4) 5
6. The condition that the equation  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ , can take the form  $aX^2 + 2hXY + bY^2 = 0$  by translating the origin to a suitable point is  
 1)  $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$   
 2)  $2fgh - af^2 - bg^2 - ch^2 = 0$   
 3)  $abc - af^2 - bg^2 - ch^2 = 0$   
 4)  $abc + 2fgh = 0$

### Rotation of axes :

7. When the angle of rotation of axes is  $\tan^{-1} 2$ , the transformed equation of  $4xy - 3x^2 = a^2$  is  
 1)  $2XY + a^2 = 0$       2)  $XY - a^2 = 0$   
 3)  $X^2 - 4Y^2 = a^2$       4)  $X^2 - 2Y^2 = a^2$
8. The angle of rotation of axes so that  $\sqrt{3}x - y + 1 = 0$  transformed as  $y = k$  is  
 1)  $\frac{\pi}{6}$       2)  $\frac{\pi}{4}$       3)  $\frac{\pi}{3}$       4)  $\frac{\pi}{2}$
9. The angle of rotation of the axes so that the equation  $x + y - 6 = 0$  may be reduced

to the form  $X = 3\sqrt{2}$  is

- 1)  $\pi/6$     2)  $\pi/4$     3)  $\pi/3$     4)  $\pi/2$
10. The coordinate axes are rotated about the origin 'O' in counter clockwise direction through an angle of  $60^\circ$ . If 'p' and 'q' are intercepts made on new axes by a straight line whose equation referred to the original axes is  $x + y = 1$ , then  $\frac{1}{p^2} + \frac{1}{q^2} =$   
 1) 2      2) 4      3) 6      4) 8
11. The coordinate axes are rotated through an angle  $22^\circ$  about the origin. If the equation  $4x^2 + 12xy + 9y^2 + 6x + 9y + 2 = 0$  changes to  $aX^2 + 2hXY + bY^2 + 2gX + 2fY + c = 0$  then value of  $\frac{g^2 - ac}{a^2 + h^2} =$   
 1)  $\frac{1}{52}$     2)  $\frac{1}{36}$     3)  $\frac{-27}{52}$     4)  $\frac{1}{40}$
12. If the axes are rotated through an angle  $\frac{\pi}{2}$  in the positive direction, the new coordinates of  $(x, y)$  are  
 1)  $(-x, -y)$       2)  $(x, -y)$   
 3)  $(y, x)$       4)  $(y, -x)$
13. If the axes are rotated through an angle  $30^\circ$  in the clock wise direction, the coordinates of  $(4, 2\sqrt{3})$  before the rotation of axes are  
 1)  $(3\sqrt{3}, 1)$       2)  $(2\sqrt{3}, 2)$   
 3)  $(2, 3)$       4)  $(\sqrt{3}, 5)$
14. If the axes are rotated through an angle  $\pi$  then the transformed equation of  $ax + by + c = 0$  is  
 1)  $aX + bY = 0$       2)  $bX - aY + c = 0$   
 3)  $bX - aY - c = 0$       4)  $aX + bY - c = 0$
15. If the transformed equation of  $xy = 0$  is  $X^2 - Y^2 = 0$ , the angle of rotation of axes is  
 1)  $\frac{\pi}{2}$     2)  $\frac{\pi}{3}$     3)  $\frac{\pi}{4}$     4)  $\frac{\pi}{6}$
16. If the axes are rotated through an

angle  $\alpha$ , the new equation of  $x \sin \alpha - y \cos \alpha = p$  is

- 1)  $X = p$                       2)  $Y = p$   
3)  $X + p = 0$                   4)  $Y + p = 0$

**LEVEL-II (H.W)- KEY**

- 1) 1              2) 1              3) 3              4) 1  
5) 2              6) 1              7) 3              8) 3  
9) 2              10) 1              11) 1              12) 4  
13) 1              14) 4              15) 3              16) 4

**LEVEL-II (H.W)- HINTS**

- New origin  $= \left( -\frac{f}{h}, -\frac{g}{h} \right)$
- New origin  $= \left( -\frac{f}{h}, -\frac{g}{h} \right) = (-2, 1)$   
put  $x = X - 2, y = Y + 1$
- $(h, k) = (-1, 2)$ , put  $x = X - 1, y = Y + 2$
- $(h, k) = (-1, 2)$  put  $x = X - 1, y = Y + 2$
- new origin  $= \left( \frac{-g}{a}, \frac{-f}{b} \right) = (2, 3)$   
put  $x = X + 2, y = Y + 3$
- Synopsis
- $\tan \theta = 2$ , Find  $\sin \theta$  and  $\cos \theta$
- $a = \sqrt{3}, b = -1, \theta = \tan^{-1} \left( \frac{-a}{b} \right)$
- $a = 1, b = 1, \theta = \tan^{-1} \left( \frac{b}{a} \right)$
- $\frac{1}{p^2} + \frac{1}{q^2} = \frac{1}{a^2} + \frac{1}{b^2}$
- $a = 4, c = 2, g = 3, h = 6$   
 $\frac{g^2 - ac}{a^2 + h^2} = \frac{1}{52}$
- Use  $X = x \cos \theta + y \sin \theta$ ,  
 $Y = -x \sin \theta + y \cos \theta$
- $\theta = -30^\circ, (X, Y) = (4, 2\sqrt{3}), (x, y) = ?$
- Use  $x = X \cos \theta - Y \sin \theta$ ,  
 $y = X \sin \theta + Y \cos \theta$
- $\theta = \frac{\pi}{4}$
- Use  $x = X \cos \alpha - Y \sin \alpha$ ,  
 $y = X \sin \alpha + Y \cos \alpha$



**LEVEL-III**

- The line joining two points A (2,0) and B(3,1) is rotated about A in anticlockwise direction through an angle  $15^\circ$ . If B goes to C then C =  
1)  $\left( \frac{4 + \sqrt{2}}{2}, \sqrt{6} \right)$               2)  $\left( \frac{6 + \sqrt{2}}{2}, \frac{\sqrt{6}}{2} \right)$   
3)  $\left( \frac{2 + \sqrt{2}}{2}, \frac{\sqrt{6}}{2} \right)$               4)  $\left( \frac{4 + \sqrt{2}}{2}, \frac{\sqrt{6}}{2} \right)$
- The line passing through (7,3), (5,1) meets the x-axis at P. If the line is rotated through an angle  $30^\circ$  in the anti clockwise direction about P then the slope of its new position is  
1)  $\sqrt{3}$     2)  $1/\sqrt{3}$     3)  $2 + \sqrt{3}$     4)  $2 - \sqrt{3}$
- The line segment joining A(3, 0), B(5, 2) is rotated about a point A in anticlockwise sense through an angle  $\frac{\pi}{4}$  and B move to C. If a point D be the reflection of C in y-axis, then D =  
1)  $(-3, 2\sqrt{2})$               2)  $(3, 2\sqrt{2})$   
3)  $(3, -2\sqrt{2})$               4)  $(3, 8\sqrt{2})$
- Shift the origin to a suitable point so that the equation  $y^2 + 4y + 8x - 2 = 0$  will not contain y and constant terms is  
1)  $(3/4, -2)$     2)  $(3/4, 2)$     3)  $(2, 3/4)$     4)  $(2, -3/4)$
- The point (3,2) undergoes the following three transformations in the order given  
(i) Reflection about the line  $y=x$   
(ii) Translation by the distance 1 unit in the positive direction of x-axis  
(iii) Rotation by an angle  $\pi/4$  about the origin in the anticlockwise direction. Then the final position of the point is (EAM-2012)  
1)  $(-\sqrt{18}, \sqrt{18})$               2)  $(-2, 3)$   
3)  $(0, \sqrt{18})$               4)  $(0, 3)$
- The origin is translated to (1,2). The point (7,5) in the old system undergoes the following transformations successively  
(i) Moves to the new point under the given translation of origin  
(ii) Translated through 2 units along the negative direction of the new X-axis

(iii) Rotated through an angle  $\frac{\pi}{4}$  about the origin of new system in the clockwise direction.

The final position of the point (7,5) is  
(EAM - 2013)

1)  $\left(\frac{9}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$       2)  $\left(\frac{7}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

3)  $\left(\frac{7}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$       4)  $\left(\frac{5}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$

7. The point (1,3) undergoes the following transformations successively

(i) Reflection with respect to the line  $y=x$   
(ii) translation through 3 units along the positive direction of the X-axis

(iii) Rotation through an angle  $\frac{\pi}{6}$  about the origin in the clockwise direction.

The final position of the point P is  
(EAM-2014)

1)  $\left(\frac{7}{\sqrt{2}}, \frac{-5}{\sqrt{2}}\right)$       2)  $\left(\frac{6+\sqrt{3}}{2}, \frac{1-6\sqrt{3}}{2}\right)$

3)  $\left(\frac{6\sqrt{3}-1}{2}, \frac{6+\sqrt{3}}{2}\right)$       4)  $\left(\frac{6\sqrt{3}+1}{2}, \frac{\sqrt{3}-6}{2}\right)$

8. If the square ABCD where A(0,0), B(2,0), C(2,2) and D (0,2) undergoes the following three transformations successively :

i)  $f_1(x, y) \rightarrow (y, x)$

ii)  $f_2(x, y) \rightarrow (x+3y, y)$

iii)  $f_3(x, y) \rightarrow \left(\frac{x-y}{2}, \frac{x+y}{2}\right)$

then the final figure is :

- 1) square      2) parallelogram  
3) rhombus      4) rectangle

9. The point (4,3) when the axes translated to the point (3,1) and then axes are rotated through  $30^\circ$  about the origin, then the new position of the point is

1)  $\left(\frac{2\sqrt{3}+1}{2}, \frac{\sqrt{3}-2}{2}\right)$

2)  $\left(\frac{\sqrt{3}+1}{2}, \frac{2\sqrt{3}+1}{2}\right)$

3)  $\left(\frac{\sqrt{3}+2}{2}, \frac{2\sqrt{3}-1}{2}\right)$       4)  $\left(\frac{\sqrt{3}-2}{2}, \frac{\sqrt{3}+1}{2}\right)$

10. If the axes are shifted to (-2,-3) and rotated through  $\frac{\pi}{4}$  then transformed equation

of  $2x^2 + 4xy - 5y^2 + 20x - 22y - 14 = 0$  is

1)  $X^2 - 14XY - 7Y^2 = 2$

2)  $X^2 - 14XY - 7Y^2 = 4$

3)  $X^2 - 14XY + 7Y^2 = 2$

4)  $X^2 + 14XY + 7Y^2 = 2$

### LEVEL-III - KEY

- 1) 4    2) 3    3) 1    4) 1    5) 3  
6) 3    7) 4    8) 2    9) 3    10) 1

### LEVEL-III - HINTS

1.  $r = \sqrt{2}$ ,  $(x_1, y_1) = (2, 0)$ ,  $\theta = 45^\circ + 15^\circ = 60^\circ$   
 $x = x_1 + r \cos \theta$

$y = y_1 + r \sin \theta$

2. Angle made by the line  $\overline{AB}$  with x-axis =  $45^\circ + 30^\circ = 75^\circ$

$\therefore$  Slope of the line in new position

$m = \tan 75^\circ = 2 + \sqrt{3}$

3.  $A = (x_1, y_1) = (3, 0)$ ,  $AB = r = 2\sqrt{2}$ ,  $\theta = 45^\circ + 45^\circ = 90^\circ$

using  $x = x_1 + r \cos \theta$ ,  $y = y_1 + r \sin \theta$

we get  $C = (3, 2\sqrt{2})$   $\therefore$  D = reflection of C  
 $(3, 2\sqrt{2})$  in Y axis =  $(-3, 2\sqrt{2})$

4.  $y^2 + 8x + 4y - 2 = 0$

New origin =  $(h, k)$

$(Y + k)^2 + 8(X + h) + 4(Y + k) - 2 = 0$

Coeff of Y = 0  $\Rightarrow 2k + 4 = 0 \Rightarrow k = -2$

Constant term = 0  $\Rightarrow k^2 + 8h + 4k - 2 = 0$

$$\Rightarrow h = \frac{3}{4} \therefore \text{New origin} = \left(\frac{3}{4}, -2\right)$$

5. Reflection of  $P(3, 2)$  with reference to  $y=x$  is  $Q(2, 3)$ . after second transformation, point  $Q(2, 3)$  becomes to  $R(3, 3)$ . after third transformation  $R$  goes to  $R^1$  such that  $OR = OR^1$  by verification option (3) is the answer.

6. After first transformation given point  $P(7, 5)$  becomes  $Q(6, 3)$ . after second transformation  $Q(6, 3)$  becomes to  $(6-2, 3) = R(4, 3)$ ,

$$OR = OR^1 = 5, \tan \theta = \frac{3}{4},$$

$$\therefore \sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5}$$

The final position of the point

$$= \left[ OR^1 \cos \left( \theta - \frac{\pi}{4} \right), OR^1 \sin \left( \theta - \frac{\pi}{4} \right) \right]$$

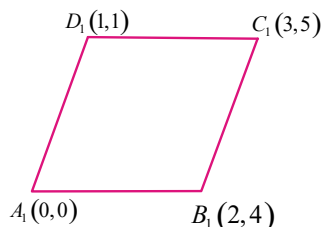
7. Reflection of  $P(1, 3)$  w.r.t the line  $y = x$  is  $Q(3, 1)$ . after second transformation,  $Q(3, 1)$  becomes to  $R(6, 1)$ ,  $OR = OR^1 = \sqrt{37}$ ,

$$\tan \theta = \frac{1}{6} \therefore \sin \theta = \frac{1}{\sqrt{37}}, \cos \theta = \frac{6}{\sqrt{37}}.$$

The final position of the point

$$= \left( OR^1 \cos \left( \theta - \frac{\pi}{6} \right), OR^1 \sin \left( \theta - \frac{\pi}{6} \right) \right)$$

8. Under the transformations  $f_1, f_2$  and  $f_3$ ,  
 $A(0, 0) \rightarrow (0, 0) \rightarrow (0, 0) \rightarrow A_1(0, 0)$   
 $B(2, 0) \rightarrow (0, 2) \rightarrow (6, 2) \rightarrow B_1(2, 4)$



$$C(2, 2) \rightarrow (2, 2) \rightarrow (8, 2) \rightarrow C_1(3, 5)$$

$$D(0, 2) \rightarrow (2, 0) \rightarrow (2, 0) \rightarrow D_1(1, 1)$$

Clearly  $A_1B_1 \parallel D_1C_1 \parallel B_1C_1$  and  $A_1C_1$  is not perpendicular to  $B_1D_1$ . therefore  $A_1B_1C_1D_1$  is parallelogram.

$$9. (x, y) = (4, 3), (h, k) = (3, 1)$$

$$\text{by using } X = (x - h) \cos \theta + (y - k) \sin \theta$$

$$10. (h, k) = (-2, -3), \theta = 45^\circ \text{ using } Y = -(x - h) \sin \theta + (y - k) \cos \theta$$

$$x - h = X \cos \theta - Y \sin \theta$$

$$y - k = X \sin \theta + Y \cos \theta$$



#### LEVEL - IV

1. **Assertion(A):** If the area of triangle formed by  $(0, 0)$ ,  $(2, 0)$ ,  $(0, 2)$  is 2 square units. Then the area of triangle on shifting the origin to a point  $(2, 3)$  is 2 units

**Reason(R):** By the change of axes area does not change

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false

4) A is false but R is false

2. **Statement I :** The point to which the origin has to be shifted to eliminate  $x$  and  $y$  terms in the equation  $a(x + \alpha)^2 + b(y + \beta)^2 = c$  is  $(-\alpha, -\beta)$

**Statement II :** The point to which the origin has to be shifted to eliminate  $x$  and  $y$  terms in  $ax^2 + by^2 + 2gx + 2fy + c = 0$  is

$$\left( \frac{-g}{a}, \frac{-f}{b} \right) \text{ Which of the above statement}$$

is true :

1) only I

2) only II

3) Both I and II

4) Neither I nor II

3. **Statement-1:** By translating the axes the equation  $xy - x + 2y = 6$  has changed to  $XY = c$  then  $c=4$   
**Statement-2:** If the axes are translated to the point  $(h,k)$ , then the equation  $f(x,y) = 0$  of a curve is transformed to  $f(X-h, Y-k) = 0$

1) Statement-1: is true, statement-2 true, statement-2 is a correct explanation for statement-1

2) Statement-1 is true, statement -2 true, statement-2 is not a correct explanation for statement-1

3) Statement-1 is true, statement -2 is false

4) statement-1 is false, statement-2 is true

4. To remove the first degree terms in the following equations origin should be shifted to the another point then calculate the new origins for List - II

List - I

List - II

- (A)  $x^2 - y^2 + 2x + 4y = 0$  1) (5, -7)  
 (B)  $4x^2 + 9y^2 - 8x + 36y + 4 = 0$  2) (1, -2)  
 (C)  $x^2 + 3y^2 - 2x + 12y + 1 = 0$  3) (-1, 2)  
 (D)  $2(x - 5)^2 + 3(y + 7)^2 = 10$  4) (-1, -2)  
 5) (-5, 7)

The correct matching is

|    | A | B | C | D |
|----|---|---|---|---|
| 1) | 4 | 2 | 2 | 5 |
| 2) | 5 | 3 | 3 | 5 |
| 3) | 3 | 2 | 2 | 1 |
| 4) | 4 | 3 | 3 | 1 |

### LEVEL-IV - KEY

- 1) 1      2) 3      3) 3      4) 3

### LEVEL-IV - HINTS

1. Area remains same

2. Synopsis

3.  $xy - x + 2y - 6 = 0$

$$h = \frac{1}{2}, g = \frac{-1}{2}, f = 1, c = -6$$

$$\text{using } 2h^2XY - 2fg + ch = 0$$

4. using new origin  $= \left( \frac{-g}{a}, \frac{-f}{b} \right)$

$$\text{We get } A = (-1, 2)$$

$$B = (1, -2), C = (1, -2), D = (5, -7)$$

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